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CS 499 Computer Science Capstone
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CS 499 Milestone Three: Enhancement Two: Algorithms and Data Structure

My only artifact is my application I am calling InvenX, which is a complete rewrite and enhancement of the Android Inventory App that I originally developed for CS 360. The original version was a native Android app built in Java and using a local SQLite database. This project initially allowed users to add, update, and delete inventory items on their device. I am transforming that app into a full-stack web application using the MEAN stack, thereby expanding the original functionality, scalability, and security of the application. I am focusing on only this project for my ePortfolio, and it has been a slow process so far.

I chose this artifact as I felt it would best show my ability and worked within a skill set that I had an interest in, as I did enjoy CS 465 and, also, it improved upon a project that I created in CS 360 that I felt lacked any real substance, despite the effort and time I put into its creation. The MEAN SPA app shows that I can adapt a basic mobile project into a web-based application using a different framework, and language. The artifact implements secure RESTful APIs, handles user authentication with JWT, and could work with a cloud database to allow for scalability. For algorithms and data structures, I improved upon, and continue to plan improvements through replacing the SQLite queries with secure CRUD operations in MongoDB, and will then turn toward implementing more efficient searching and sorting features for inventory management UI user experience.

Going back to my initial plans from Module One, some of the course outcomes are already met, and some are still a work in progress. I have, by my research, rewrote the backend and database, implemented user authentication securely, and followed security practices through JWT tokens stored in the browser. I am going to be honest, I have not fully grasped implementation of the searching, but I believe I should be able to implement this on the front end, so the back end should be nearly complete. Any of the advanced search and sorting in the inventory rely on me completing the CRUD operations and frontend UI first, so realistically, nothing has been performed for Algorithms and Data Structure enhancement.

Thus far, I have learned a lot about full-stack development, especially in routing, security, and allowing the backend, and frontend to work together. Another important point in design and testing that I have learned is that the backend APIs needs to be fully implemented and completed prior to building out the UI. The biggest challenges have been making all the different parts of the MEAN stack communicate smoothly, especially when I was still testing and debugging with Postman. Also, I am gaining a lot of experience in reading errors in the developer area of a web browser, and how to troubleshoot error messages. I do feel a little more confident, and I feel more aware of how to tackle problems before they occur when connecting multiple layers of an application.

I will say I basically wasted around 12 hours of my time this past week attempting to implement objects and structures that I did not initially intend in the project. I had a brilliant idea while at work to allow users to transverse to a screen with their information, and then update and change information, including passwords. I did not realize how extensive this undertaking would be during development. After around 4 hours of YouTube videos and many hours of debugging, I removed this attempted additional feature, and decide to utilize the '/me' route to simply add a

“Welcome, {{ username }}!” to the inventory screen. This should suffice for the individuality of the application, but I think if I do continue to improve this application after school is completed, that is one of my undertakings.